Q1)Answer:

1. Binomial Tree

A binomial tree is a recursively defined tree structure used as the basic building block of a binomial heap. It is denoted by B^k, where k represents the order of the tree.

Properties of a Binomial Tree

1. A binomial tree of order 0, B_0, consists of a single node.
2. A binomial tree of order k, B_k, is formed by linking two binomial trees of order k-1, making one tree the leftmost child of the other.
3. The number of nodes in B_k is 2^k.
4. The maximum height of B_k is k.
5. For B_k , the number of nodes at depth i are ${}^kC_i$ (k combination i) for i = 0,1,2,3.....(root marked as i = 0)
6. The root has degree k , greater than any other node.
7. If I the children of the root are numbered from Left to Right by k-1 , k-2 , ... , 0 then child I is the root of subtree B_i .

2. Binomial Heap

A binomial heap is a collection of binomial trees that satisfies two important properties: the binomial-tree structure property and the heap-order property. It is used to implement priority queues efficiently.

Properties of a Binomial Heap(H)

1. Every tree in the heap is a binomial tree.
2. Each binomial heap obeys the min/max heap property.
3. For a non-negative integer k, there is at most one(only one) binomial tree B_k in H.
4. The roots of the binomial trees are maintained in a linked list, generally in increasing order of degree.
5. A binomial heap containing n nodes has at most floor($\log_2 n$) +1 trees.

Q2) Answer: c

A binomial heap containing n nodes has at most floor($\log_2 n$) +1 trees.

Q3) Answer: b

This could be easily verified by looking at the structure of a binomial heap.

Q4)Answer: c

The function return minimum value in the heap_Array which is equal to the root value of the heap.

Q5)Answer: a

1. If x is not a root

The sibling of x has a smaller degree than x. Because in a binomial tree, children of a node are arranged in decreasing order of degree from left to right. A non-root node's sibling appears to its right, so its sibling has a smaller degree.

2. If x is a root

The sibling of x has a larger degree than x. Because the roots in a binomial heap's root list are arranged in increasing order of degree from left to right. Therefore, the next root (the sibling) has a larger degree.

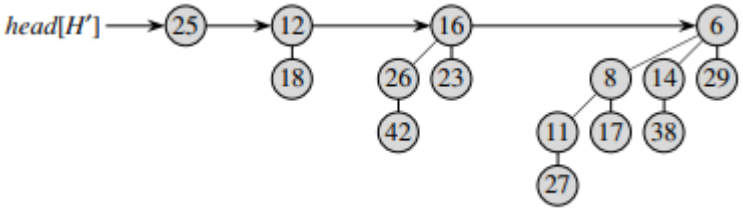
Note: In the CLRS representation of a binomial tree, each node has:

- parent[x]: its parent

- child[x]: its leftmost child
- sibling[x]: its immediate right sibling, not just any sibling

Q6) Answer: a
Convert 13 to binary:
13=> 1101 = 8+4+1
Therefore, the heap contains B_0, B_2, B_3 , since the set bits (i.e. 1's) correspond to those binomial trees.

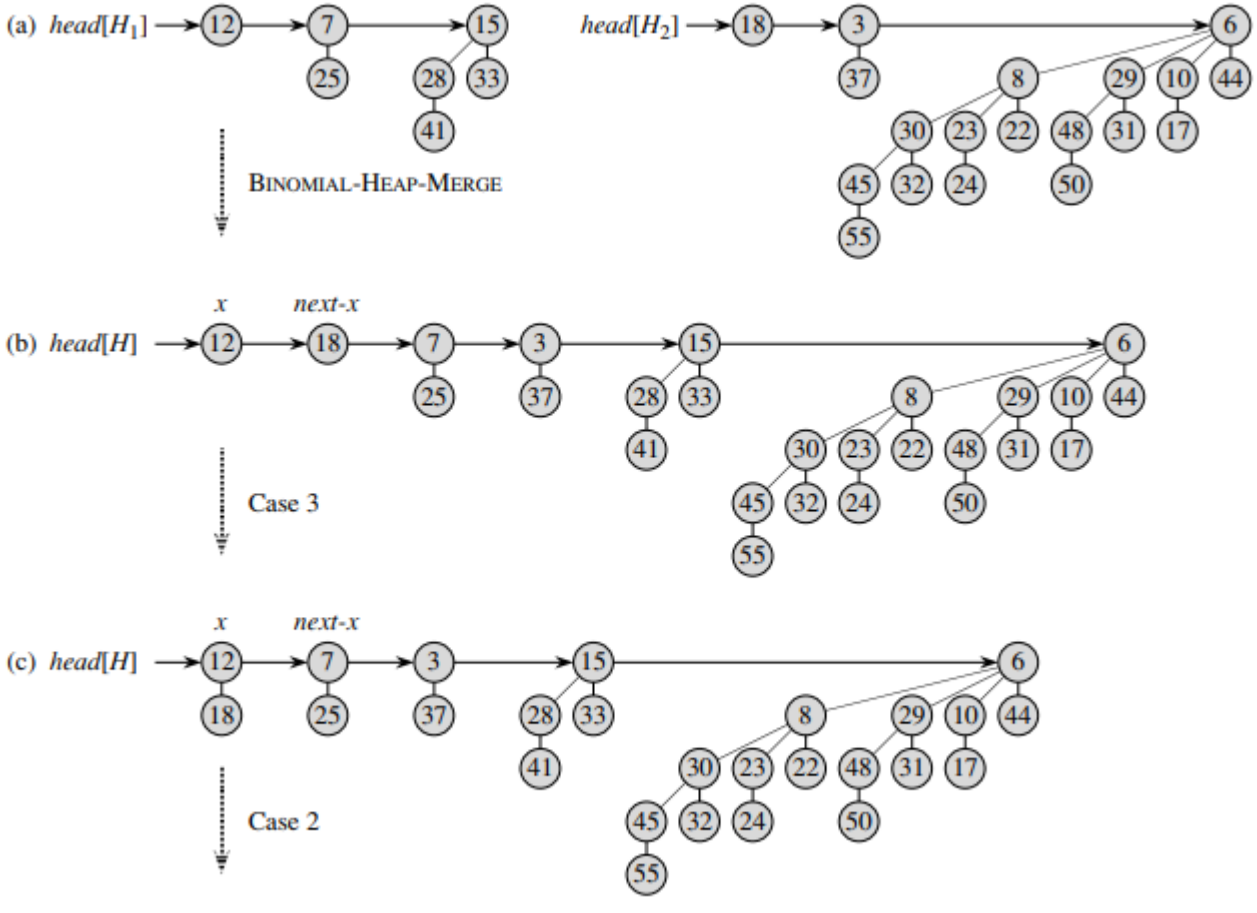
Q7) Answer:

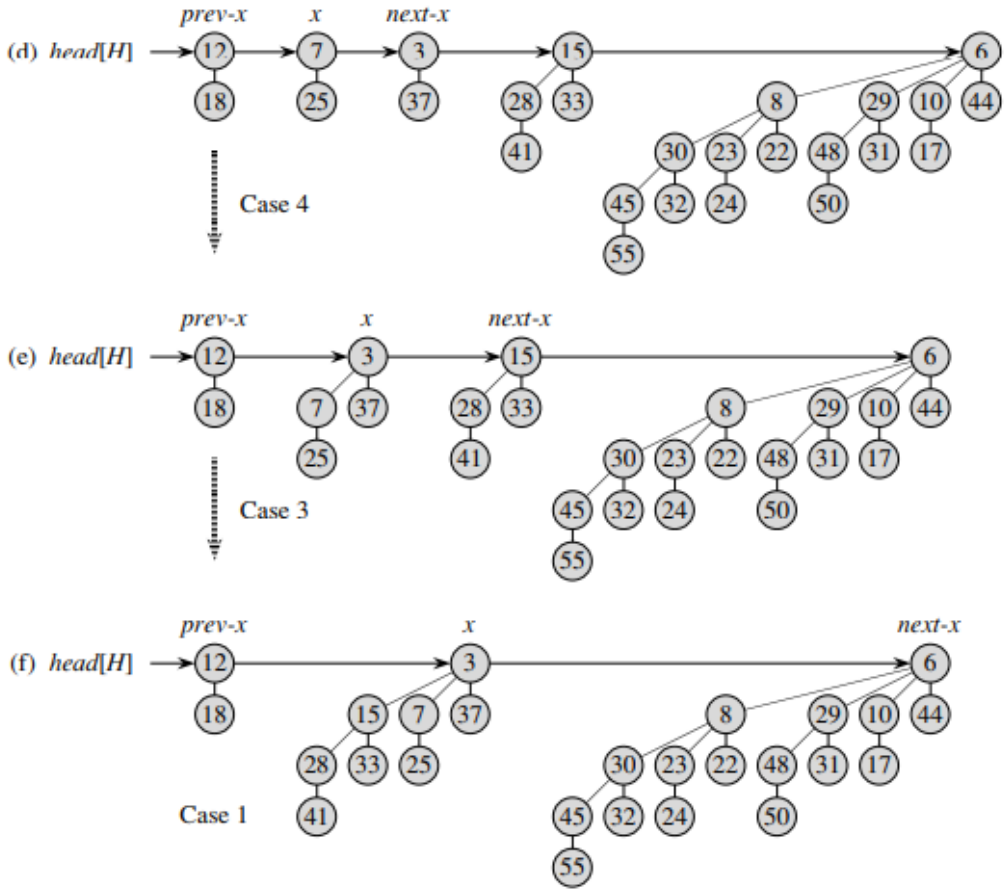


One possible configuration of a min binomial heap. The values inside the nodes may differ, provided the min-heap property is not violated; however, the structure of the binomial trees remains the same.

Q8) Answer: 3,2,1,0
Convert 15 into binary:
15= $(1111)_2$ =8+4+2+1
The heap contains B_3, B_2, B_1 and B_0
Thus, the degrees in decreasing order are:
3,2,1,0

Q9) First see the union operation in binomial heap.





Q10) First see the extract min operation in binomial heap

